

- of stocks the previous day, and recommends which of his equity holdings the user can sell that day for a profit of 10 per cent or more. Your recommendation should contain the company name, number of shares to sell, the sale price and the percentage of profit that the user will make if he sells at that price. What is the time complexity of your algorithm, assuming that you are given N transactions in total, corresponding to M different companies?
11. Consider the above question. Stock markets are quite volatile and the price of company stocks can fluctuate considerably during the day. You are now told that instead of using the previous day's closing price, your application should read in real-time the current price of the stocks that the user holds, and then recommend which stocks the user can sell at that point of time for a 10 per cent profit. Assume that you have stock market prices available as a real-time feed at every 60 second interval. How will you modify your solution for question (10) to satisfy this criterion?
 12. You are given a search query and you can use any of your favourite Web search engines to find the query results. Web search engines use their own ranking algorithm to rank the query search results. However, you are asked to apply a filter on top of the search engine query results such that the results are ordered temporally. For example, consider an example query of the form: 'Donald Trump US Presidential election'. Given a set of search engine query results for this query which will contain news articles, events, etc, you need to sort them temporally. For instance, a query result link on the news story from July 2015 describing Trump's intention to run should be ranked before the news story featuring his 2016 New Year wishes to the American electorate. Essentially, your task is to organise the search engine query results on a timeline. What are the challenges here?
 13. You are given four strings S_1 , S_2 , S_3 and S_4 . S_1 is a large string and S_2 , S_3 , and S_4 are comparatively small strings. You are asked to find the smallest length sub-string 't' of S_1 such that it contains S_2 , S_3 and S_4 as its sub-strings. For example, given string S_1 'BytheriversofBabylon', and S_2 =er, S_3 =of and S_4 =so, your function should return 'ersof'. How will you modify your solution if you are now told that the ordering of S_2 , S_3 , and S_4 should be maintained in the sub-string 't' (i.e., S_2 should appear before S_3 , and S_3 should appear before S_4 in sub-string 't')?
 14. When we input a query into a search engine such as Google, we get an 'auto complete' suggestion for it based on past queries that have been tried on the search engine. Can you explain how you will design an auto-complete mechanism that will scale well across the millions of queries that can happen concurrently at a time? What is the data structure that you will use which can store past queries and automatically provide suggestions? Will you store the complete query as is or will you break it into its constituent words? Given that scalability and correctness is key for this problem, can you explain how the choice of your data structure will address both?
 15. You are given a set of documents and asked to perform their clustering based on document level similarity. You are familiar with different clustering algorithms such as 'kmeans', so you are confident that you can do the clustering. However, the interviewer now adds a second part to the problem where he wants you to label the clusters. How will you generate labelled clusters? What are the means of generating meaningful labels for the clusters? Can the clusters be labelled post the clustering process, prior to it, or during the cluster process itself? Describe your choice of the solution and why you made it.
 16. You are given an array A of N integers. You are also given a number X . You are asked to find the element in A that is closest to X . You are given the constraint that you should not sort the array A . (a) Can you write a C program to solve this problem? (b) Now you are told that the set S consists of floating point values instead of integers. How will you modify your solution for (a) to solve (b)?
 17. Given a classifier, can you define the terms 'sensitivity' and 'specificity' of your system? Let us assume that you have built a classifier that can predict whether a tumor is benign or malignant based on certain tumor characteristics. Your intention is to build a classifier that can detect all true malignant tumors, even if some of the benign tumors get falsely reported as malignant. Which of the two metrics — sensitivity or specificity — should you try to improve in this case? Now assume that you have been asked to use a classifier to predict whether a student will pass the examination or not. Your system should be designed such that it does not report a failing student as passing even if it mispredicts certain passing students as potential failure students. In that case, which of these two metrics should you try to improve?
 18. What is a maximum margin classifier? Can you give an example? What is the advantage of using this classifier?
 19. Graph neural networks are becoming popular in machine learning and natural language processing. Can you explain the details of a graph convolutional network (GCN)? How is it different from a standard convolutional network (CNN)?
 20. While transfer learning has become very popular and common in both computer vision and NLP, can you list some scenarios where it may not be useful and you may need to train a model from scratch?
- Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers a very happy new year 2022! Stay healthy and stay safe. 

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